

Bayes 1

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March 2026

1 The beta-binomial model

1.1 Compute the prior mean of θ and provide a 95%-credibility interval for the prior.

We have that:

$$\theta \sim \text{Beta}(a_0, b_0), \quad a_0 = b_0 = 1$$

The mean of this beta distribution can then be calculated as:

$$\mathbb{E}[\theta] = \frac{a_0}{a_0 + b_0} = \frac{1}{2}$$

We also need to provide a 95%-credibility interval for the prior distribution. According to Murphy, "we define a $100(1 - \alpha)\%$ credible interval to be a (contiguous) region $C = (l, u)$ (standing for lower and upper) which contains $1 - \alpha$ of the posterior probability mass". There can be multiple such intervals, so we use the central interval, i.e. an interval where there is $\alpha/2$ mass in each tail. Normally, this calculation is cumbersome and requires the inverse CDF or iterative definite integrals, but we realise that $\text{Beta}(a_0, b_0)$ is the uniform distribution, a so-called "flat prior", meaning that the probability density is 1 inside the whole domain from 0 to 1. Therefore, the 95%-credibility interval must be:

$$(l, u)_\theta = \left(\frac{\alpha}{2}, \frac{1 - \alpha}{2} \right) = (0.025, 0.975)$$

1.2 Compute the posterior mean of θ and provide a 95%-credibility interval for the posterior.

We know from week 2 of this course that if:

$$p(\theta) = \text{Beta}(a_0, b_0)$$

$$p(y|\theta) = \binom{N}{y} \theta^y (1 - \theta)^{N-y}$$

Then, we can use Bayes rule to calculate the following posterior:

$$p(\theta|y) = \text{Beta}(y + a_0, N - y + b_0)$$

Given $N = 115$, $y = 4$, and $a_0 = b_0 = 1$, we can calculate the posterior mean as:

$$\mathbb{E}[\theta|y] = \frac{y + a_0}{N - y + b_0} = \frac{4 + 1}{115 - 4 + 1} = \frac{5}{113} \approx 0.044$$

We will calculate the 95% credibility interval using `scipy.stats.beta.interval`:

$$(l, u)_{\theta|y} = (0.014, 0.086)$$

1.3 Compute and plot the posterior predictive distribution for y^* given $y = 4$.

From week 2, we know that the posterior predictive distribution can be written as:

$$p(y^* = k|y) = \binom{N^*}{k} \frac{B(\alpha + k, \beta + N^* - k)}{B(\alpha, \beta)}, \quad \alpha = y + a_0, \quad \beta = N - y + b_0$$

Where B is the Beta function. In our case, we have that $N^* = 20$, $y = 4$, $N = 115$, and $a_0 = b_0 = 1$. There, the expression simplifies to:

$$p(y^* = k|y) = \binom{20}{k} \frac{B(5 + k, 133 - k)}{B(5, 113)}$$

To plot this distribution, we simply input all possible values of k , i.e. $k = 0, 1, \dots, 20$ and plot the probability densities $p(y^* = k|y)$ as a function of k . The probability densities are visualized in Figure 1.

1.4 What is the posterior predictive probability that at least one of the next $N^* = 20$ customers will make a purchase?

This is the same as asking:

$$p(y^* \geq 1|y)$$

Or alternatively:

$$1 - p(y^* < 1|y) = 1 - p(y^* = 0|y)$$

We can plug in the values that we calculated in the prior exercise to get that:

$$p(y^* \geq 1|y) = 1 - p(y^* = 0|y) \approx 1 - 0.45 = 0.55$$

1.5 Compute mean and variance of the posterior predictive distribution for y^*

The posterior predictive distribution is, in this case, a discrete probability. Therefore, the mean and variance can be calculated as:

$$\mathbb{E}[y^*|y] = \sum k \cdot p(y^* = k|y)$$

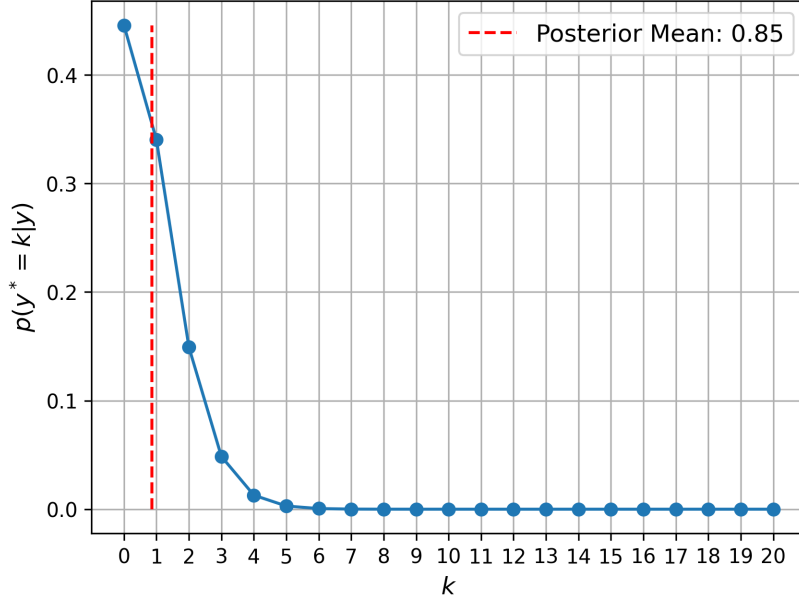


Figure 1: The posterior predictive distribution.

And the variance can be calculated as:

$$\text{Var}[y^*|y] = \sum (k - \mathbb{E}[y^*|y])^2 p(y^* = k|y)$$

In both cases, the sum runs over all values in the domain of the probability distribution, i.e. $k = 0, 1, \dots, 20$ in our case. We can use the values from exercise 1.3 and plug them into the formulas to obtain:

$$\mathbb{E}[y^*|y] \approx 0.85$$

$$\text{Var}[y^*|y] \approx 0.95$$

2 Linear Gaussian systems

2.1 Determine the distribution $p(y)$

We have that:

$$\begin{aligned} p(\mathbf{z}_1) &= \mathcal{N}(\mathbf{0}, v\mathbf{I}) \\ p(\mathbf{z}_2|\mathbf{z}_1) &= \mathcal{N}(\mathbf{z}_1, v\mathbf{I}) \\ p(y|\mathbf{z}_2) &= \mathcal{N}(\mathbf{a}^T \mathbf{z}_2, \sigma^2) \end{aligned}$$

We want to calculate $p(y)$. We do this by factoring the probability in the following way:

$$p(y, \mathbf{z}_2) = p(y|\mathbf{z}_2)p(\mathbf{z}_2) \Rightarrow p(y) = \int p(y|\mathbf{z}_2)p(\mathbf{z}_2) d\mathbf{z}_2$$

We don't know the distribution $p(\mathbf{z}_2)$, so we will deduce this first. One way to do this is to use equation (3.38) in the book by Murphy. The general equation says that:

$$p(y) = \int \mathcal{N}(z|\boldsymbol{\mu}_z, \Sigma_z) \mathcal{N}(y|\mathbf{W}\mathbf{z} + \mathbf{b}, \Sigma_y) d\mathbf{z} = \mathcal{N}(y|\mathbf{W}\boldsymbol{\mu}_z + \mathbf{b}, \Sigma_y + \mathbf{W}\Sigma_z\mathbf{W}^T)$$

This is the general equation, as seen in the book. For our purposes, we will instead use the equation to deduce an expression for $p(\mathbf{z}_2)$. Therefore, we have to match the terms from the general equation with our terms, and we find that: $\boldsymbol{\mu}_z = \mathbf{0}$, $\Sigma_z = v\mathbf{I}$, $\mathbf{W} = 1$, $\mathbf{b} = \mathbf{0}$, and $\Sigma_y = v\mathbf{I}$. In this case, the expression simplifies to:

$$p(\mathbf{z}_2) = \mathcal{N}(\mathbf{0}, v\mathbf{I} + v\mathbf{I}) = \mathcal{N}(\mathbf{0}, 2v\mathbf{I})$$

Now, having obtained an expression for $p(\mathbf{z}_2)$, we can write $p(y)$ as:

$$p(y) = \int \mathcal{N}(\mathbf{a}^T \mathbf{z}_2, \sigma^2) \mathcal{N}(\mathbf{0}, 2v\mathbf{I}) d\mathbf{z}_2$$

Once again, we recognize this integral in equation (3.38) in the book by Murphy. Therefore, once again, we simply match the terms and see that, in this specific case, the matching terms are: $\mathbf{W} = \mathbf{a}^T$, $\mathbf{b} = \mathbf{0}$, $\Sigma_y = \sigma^2$, $\boldsymbol{\mu}_z = \mathbf{0}$, and $\Sigma_z = 2v\mathbf{I}$. Inserting these values, we get:

$$p(y) = \mathcal{N}(y|\mathbf{a}^T \mathbf{0} + \mathbf{0}, \sigma^2 + \mathbf{a}^T (2v\mathbf{I}) \mathbf{a}) = \mathcal{N}(\mathbf{0}, \sigma^2 + 2v\mathbf{a}^T \mathbf{a})$$

2.2 Determine the distribution $p(y, \mathbf{z}_2|\mathbf{z}_1)$

We note that this is a 3 dimensional gaussian distribution. We will use the following fact:

$$p(y, \mathbf{z}_2|\mathbf{z}_1)p(\mathbf{z}_1) = p(y, \mathbf{z}_2, \mathbf{z}_1) \Rightarrow p(y, \mathbf{z}_2|\mathbf{z}_1) = \frac{p(y, \mathbf{z}_1, \mathbf{z}_2)}{p(\mathbf{z}_1)}$$

Next, we see that we rewrite $p(y, \mathbf{z}_1, \mathbf{z}_2)$ using a graphical model, i.e. we can use the fact that y depends only on $\mathbf{z}_2$ (this is also given in the exercise):

$$p(y, \mathbf{z}_1, \mathbf{z}_2) = p(y|\mathbf{z}_2)p(\mathbf{z}_2|\mathbf{z}_1)p(\mathbf{z}_1)$$

Inserting this, we get:

$$p(y, \mathbf{z}_2|\mathbf{z}_1) = \frac{p(y|\mathbf{z}_2)p(\mathbf{z}_2|\mathbf{z}_1)p(\mathbf{z}_1)}{p(\mathbf{z}_1)} = p(y|\mathbf{z}_2)p(\mathbf{z}_2|\mathbf{z}_1)$$

Now, we use equation (3.33) through (3.36) in the book by Murphy which says the following:

$$\text{if } p(\mathbf{z}) = \mathcal{N}(\mathbf{z}|\boldsymbol{\mu}_z, \Sigma_z) \quad \text{and} \quad p(y|\mathbf{z}) = \mathcal{N}(y|\mathbf{W}\mathbf{z} + \mathbf{b}, \Sigma_y)$$

$$\text{then } p(\mathbf{z}, y) = \mathcal{N}(\boldsymbol{\mu}, \Sigma) \quad \text{where} \quad \boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_z \\ \mathbf{W}\boldsymbol{\mu}_z + \mathbf{b} \end{pmatrix} \quad \Sigma = \begin{pmatrix} \Sigma_z & \Sigma_z \mathbf{W}^T \\ \mathbf{W}\Sigma_z & \Sigma_y + \mathbf{W}\Sigma_z \mathbf{W}^T \end{pmatrix}$$

Now, this doesn't perfectly match our expression: the above equation involves $p(\mathbf{z})$ and we have a conditional probability, namely $p(\mathbf{z}_2|\mathbf{z}_1)$. However, we can still use the equation since y doesn't explicitly depend on $\mathbf{z}_1$. Once again, we match the terms and find that $\boldsymbol{\mu}_z = \mathbf{z}_1$, $\mathbf{W} = \mathbf{a}^T$, $\mathbf{b} = \mathbf{0}$, $\Sigma_z = v\mathbf{I}$, and $\Sigma_y = \sigma^2$. Inserting this, we get:

$$p(\mathbf{z}_2, y|\mathbf{z}_1) = \mathcal{N}\left(\begin{pmatrix} \mathbf{z}_2 \\ \mathbf{a}^T \mathbf{z}_2 \end{pmatrix}, \begin{pmatrix} v\mathbf{I} & v\mathbf{I}\mathbf{a} \\ v\mathbf{a}^T \mathbf{I} & \sigma^2 + v\mathbf{a}^T \mathbf{a} \end{pmatrix}\right)$$

2.3 Determine the distribution $p(y|\mathbf{z}_1)$

We already have the distribution $p(y, \mathbf{z}_2|\mathbf{z}_1)$. Therefore, we can calculate $p(y|\mathbf{z}_1)$ as:

$$p(y|\mathbf{z}_1) = \int p(y, \mathbf{z}_2|\mathbf{z}_1) d\mathbf{z}_2$$

This is also called "marginalization". Since we are dealing with a Gaussian distribution, we can simply look at the mean and the covariance matrix of the distribution $p(y, \mathbf{z}_2|\mathbf{z}_1)$ and picking the corresponding rows and columns that doesn't depend on $\mathbf{z}_2$. Therefore, we get:

$$p(y|\mathbf{z}_1) = \mathcal{N}(\mathbf{a}^T \mathbf{z}_1, \sigma^2 + v\mathbf{a}^T \mathbf{a})$$

2.4 Determine the distribution $p(\mathbf{z}_1|y)$

This distribution is also known as the posterior. Here, we can use Bayes:

$$p(\mathbf{z}_1|y) = \frac{p(y|\mathbf{z}_1)p(\mathbf{z}_1)}{p(y)}$$

We know all of these distributions: $p(\mathbf{z}_1)$ are given in the exercise, and we have earlier deduced the expressions $p(y)$ and $p(\mathbf{z}_1|y)$. Normally, one would have

to substitute the distributions into the formula for Bayes and completing the square to determine the new mean and variance. However, since our system is a Linear Gaussian system, we can also use equation 3.37 in the book by Murphy which says the following:

$$p(\mathbf{z}|y) = \mathcal{N}(\mathbf{z}|\boldsymbol{\mu}_{z|y}, \Sigma_{z|y}) \quad \text{where}$$

$$\Sigma_{z|y}^{-1} = \Sigma_z^{-1} + \mathbf{W}^T \Sigma_{y|z}^{-1} \mathbf{W} \quad \text{and} \quad \boldsymbol{\mu}_{z|y} = \Sigma_{z|y} [\mathbf{W}^T \Sigma_{y|z}^{-1} (y - \mathbf{b}) + \Sigma_z^{-1} \boldsymbol{\mu}_z]$$

Once again, we can identify the terms as: $\Sigma_z = v\mathbf{I}$, $\Sigma_{y|z} = \sigma^2 + v\mathbf{a}^T \mathbf{a}$, $\mathbf{W} = \mathbf{a}^T$, $\mathbf{b} = \mathbf{0}$, and $\boldsymbol{\mu}_z = \mathbf{0}$. Inserting these values, we get:

$$\Sigma_{z|y}^{-1} = (v\mathbf{I})^{-1} + \mathbf{a}(\sigma^2 + v\mathbf{a}^T \mathbf{a})^{-1} \mathbf{a}^T = \frac{1}{v} \mathbf{I} + \frac{\mathbf{a}\mathbf{a}^T}{\sigma^2 + v\mathbf{a}^T \mathbf{a}}$$

Which means that:

$$\Sigma_{z|y} = \left(\frac{1}{v} \mathbf{I} + \frac{\mathbf{a}\mathbf{a}^T}{\sigma^2 + v\mathbf{a}^T \mathbf{a}} \right)^{-1}$$

And:

$$\boldsymbol{\mu}_{z|y} = \Sigma_{z|y} \left[\frac{y}{\sigma^2 + v\mathbf{a}^T \mathbf{a}} \mathbf{a} \right]$$

3 A conjugate model for count data

This section is highly computational and written by different person that intends to be far more terse. It will therefore mainly contain mathematical derivation and little text.

3.1 Determine the joint distribution $p(\mathbf{y}, \lambda)$

We assume that the likelihood of each observation $p(y_i|\lambda)$ is conditionally independent. For notational convenience I will often omit the index and limits of sum and product symbols. In this case, they are assumed to sum over i from 1 to N , i.e. $\sum$ means $\sum_{i=1}^N$.

$$\begin{aligned} p(y_i|\lambda) &= \frac{\lambda^{y_i} e^{-\lambda}}{y_i!} \\ p(\mathbf{y}, \lambda) &= p(\mathbf{y}|\lambda)p(\lambda) = \left(\prod_{i=1}^N p(y_i|\lambda) \right) p(\lambda) \\ &= \frac{\lambda^{\sum y_i} e^{-N\lambda}}{\prod y_i!} \cdot \frac{b_0^{a_0}}{\Gamma(a_0)} \lambda^{a_0-1} e^{-b_0\lambda} \\ &= \frac{\lambda^{a_0-1+\sum y_i} e^{-(b_0+N)\lambda} b_0^{a_0}}{\Gamma(a_0) \prod y_i!} \end{aligned}$$

3.2 Functional form of a Gamma distribution

We simply compute the natural logarithm of the density of a gamma distribution as given in the assignment.

$$\begin{aligned} \log p(\lambda|a, b) &= \log \left(\frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda} \right) = a \log b + (a-1) \log \lambda - b\lambda - \ln \Gamma(a) \\ &\propto (a-1) \log \lambda - b\lambda \end{aligned}$$

3.3 Derive the analytical expression for the posterior distribution $p(\lambda|\mathbf{y})$

We could use the above calculation and instead compute all probabilities in the log domain, but these calculations are straightforward enough to warrant

sticking with the probability domain.

$$\begin{aligned}
p(\lambda|\mathbf{y}) &= \frac{p(\mathbf{y}, \lambda)}{p(\mathbf{y})} \\
p(\mathbf{y}) &= \int_0^\infty p(\mathbf{y}, \lambda) d\lambda = \int_0^\infty \frac{\lambda^{a_0-1+\sum y_i} e^{-(b_0+N)\lambda} b_0^{a_0}}{\Gamma(a_0) \prod y_i!} d\lambda \\
&= \frac{b_0^{a_0}}{\Gamma(a_0) \prod y_i!} \int_0^\infty \lambda^{a_0-1+\sum y_i} e^{-(b_0+N)\lambda} d\lambda \\
&= \frac{b_0^{a_0}}{\Gamma(a_0) \prod y_i!} \cdot \frac{\Gamma(a_0 + \sum y_i)}{(b_0 + N)^{a_0+\sum y_i}}
\end{aligned}$$

In the last step we recognize that the integrand is the integral of an unnormalized Gamma distribution over the entire domain of λ . Therefore the result of the integral is the reciprocal of the corresponding gamma distribution normalization constant.

Now all that is left is to compute the posterior.

$$\begin{aligned}
p(\lambda|\mathbf{y}) &= \frac{\lambda^{a_0-1+\sum y_i} e^{-(b_0+N)\lambda} b_0^{a_0}}{\Gamma(a_0) \prod y_i!} \cdot \left(\frac{b_0^{a_0}}{\Gamma(a_0) \prod y_i!} \cdot \frac{\Gamma(a_0 + \sum y_i)}{(b_0 + N)^{a_0+\sum y_i}} \right)^{-1} \\
&= \frac{\lambda^{a_0-1+\sum y_i} e^{-(b_0+N)\lambda} (b_0 + N)^{a_0+\sum y_i}}{\Gamma(a_0 + \sum y_i)} \\
&= \text{Gamma}(a_0 + \sum y_i, b_0 + N)
\end{aligned}$$

3.4 Determine the posterior distribution for λ given data

Inserting the constants from the exercise into the above equation gives the distribution:

$$\begin{aligned}
\text{Gamma}(a_0 + \sum y_i, b_0 + N) &= \text{Gamma}(43, 5.1) \\
\lambda &\sim \text{Gamma}(43, 5.1) \\
\mathbb{E}[\lambda] &= \frac{43}{5.1} = 8.43
\end{aligned}$$

The posterior mean differs from the prior mean $\lambda \sim \text{Gamma}(1, 0.1)$, $\mathbb{E}[\lambda] = 10$ but this hardly captures how significantly the distributions differ.

3.5 Plot $p(\lambda)$ and $p(\lambda|\mathbf{y})$ for $\lambda \in [0, 30]$.

Plotting the posterior and prior distributions we see pronounced differences.

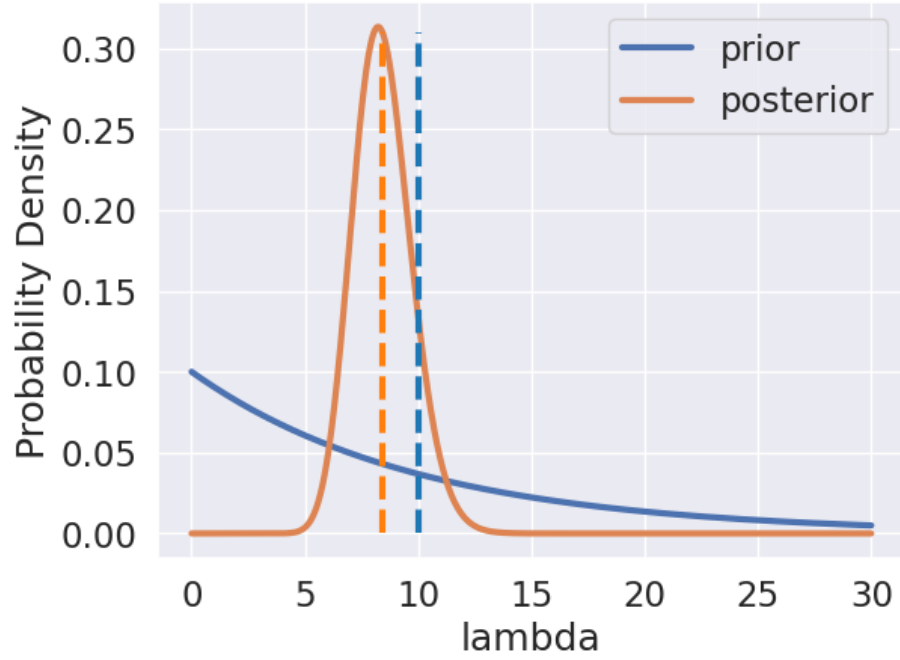


Figure 2: Prior and posterior gamma distribution for the values given in exercise 4.

In fact these gamma distributions are Erlang distributions as the shape parameters are integers. Interpreting them as Erlang distributions we can see that the prior distribution is simply an exponential distribution with rate $\frac{1}{10}$, i.e. memoryless exponential decay. The posterior, meanwhile, can be seen as the sum of 43 i.i.d. exponential distributions with rate 5.1. For $n = 43$ exponential distributions we begin to see the effect of the central limit theorem in action and notice a much more defined gaussian like density.